ModelIt!

Teacher's Guide

Can We Actually Live on Mars?

Modeling Survival in the Most Hostile Environment Humans Have Ever Faced

Quick Reference

Grade Level	Grade 09L1
NGSS Standard	HS-ESS1-2, HS-ETS1-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS1-2, HS-ETS1-2

Construct an explanation of the Big Bang theory based on astronomical evidence including light spectra, motion of distant galaxies, and composition of matter in the universe. Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct explanations of Mars survival challenges and design habitat solutions based on computational model evidence.
Disciplinary Core Idea	**ESS1.B Earth and the Solar System / ETS1.C Optimizing the Design Solution** — Students analyze the conditions on Mars that differ from Earth and engineer solutions to sustain human life.
Crosscutting Concept	**Systems and System Models** — Students model Mars habitat as a system where multiple interacting components must simultaneously maintain conditions for human survival.

CAST Testing Alignment

MULTIPLE CHOICE:

A Mars habitat experiences a slow leak that reduces atmospheric pressure by 20% over three days. According to your model, which component would be affected FIRST?

CONSTRUCTED RESPONSE:

Using your model, explain why Mars colonization requires solving all five survival factors simultaneously. What happens when even one factor drops below the critical threshold?

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Mars is the most Earth-like planet in our solar system — and it's still incredibly hostile to human life. With less than 1% of Earth's atmospheric pressure, average temperatures of -60°C, no breathable oxygen, lethal radiation, and no liquid surface water, Mars represents the ultimate engineering challenge: can humans recreate the conditions for life on a world that has none?

Mars vs. Earth

Mars has 38% of Earth's gravity, a thin CO2 atmosphere (95.3% CO2) at 0.6% of Earth's pressure, no global magnetic field, and surface temperatures from -125°C to 20°C. A human standing unprotected on Mars would simultaneously suffocate, freeze, be irradiated, and have their blood boil due to low pressure — all within about 15 seconds.

Atmospheric Pressure Challenge

Earth's atmospheric pressure is 101.3 kPa. Mars surface pressure is 0.6 kPa. Liquid water requires at least 6.1 kPa; human breathing requires about 47.3 kPa of pressure with appropriate oxygen concentration. Any habitat must maintain this pressure differential against the near-vacuum of Mars — and a breach could be catastrophic.

Radiation on Mars

Without Earth's protective magnetic field and thick atmosphere, Mars receives about 2.5 times the radiation astronauts experience on the ISS. Over a two-year stay, colonists would receive enough radiation to significantly increase cancer risk. Shielding options include burying habitats under regolith, using water walls, or developing advanced materials.

Water and Food Production

Mars has significant water ice at its poles and underground. Extracting and purifying this water is essential for drinking, agriculture, and producing oxygen through electrolysis. Growing food on Mars requires controlled environments with artificial lighting, recycled water, and processed regolith — a system that must be virtually 100% efficient since resupply from Earth takes 6-9 months.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify atmospheric pressure, radiation exposure, water availability, temperature, and crop viability as the critical components of the Mars survival system.

E **Establish Relationships**
Students determine that all four external factors must be within acceptable ranges simultaneously for crop viability to be maintained — and that any single failure cascades through the system.

V **Visualize & Model**
Students build a computational model showing how Mars habitat conditions interact to determine the feasibility of sustained human presence.

E **Evaluate Outcomes**
Students run optimal, pressure breach, and water crisis scenarios to test their model and identify critical failure thresholds.

R **Revise & Extend**
Students add energy production, oxygen generation, or waste recycling to model more realistic habitat dynamics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Elon Musk says we'll have people on Mars by 2030. But here's what he doesn't talk about: how do you keep them alive?"*

Watch For

Show a brief video of Mars surface conditions. Let students react to the -60°C average temperature.

Slide 2: Learning Objectives

3 min

Say: *"Today you're engineering a solution to the hardest survival problem in human history."*

Watch For

Have students read objectives. Pre-teach 'atmospheric pressure' and 'bioregenerative life support.'

Slide 3: Big Question

3 min

Say: *"Forget the movies. On real Mars, you'd be dead in 15 seconds without protection. How do we fix that?"*

Watch For

Earth vs. Mars comparison chart. Students identify what's missing on Mars for human survival.

Slide 4: LEVER Framework

2 min

Say: *"We're building a model of a Mars habitat to figure out what actually keeps people alive — and what kills them."*

Watch For

Preview LEVER steps. Emphasize that this is a SYSTEM problem — everything is connected.

Slide 5: Activity 1: Components

8 min

Say: *"What are the non-negotiable requirements for human survival? Which ones does Mars provide?"*

Watch For

Guide sorting. All four environmental factors are external — Mars controls them, not us.

Slide 6: Activity 2: Connections

8 min

Say: *"If pressure drops, what happens to water? To crops? To the people inside?"*

Watch For

Help students trace cascade failures. Emphasize that all paths lead to crop viability.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's break our habitat and see what fails first — and fastest."*

Watch For

Students predict failure cascades, then run scenarios. Compare pressure breach vs. water crisis.

Slide 8: Discoveries

5 min

Say: *"So is Mars colonization actually possible? What does your model say about the engineering challenges?"*

Watch For

Facilitate evidence-based debate. Which challenge is hardest? What would you solve first?

Slide 9: STEM Challenge

12 min

Say: *"NASA wants your design. Build a habitat that solves ALL five problems for six people for one year."*

Watch For

Groups design habitat modules. Must address all five factors with failure contingencies.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atmospheric Pressure, Radiation Exposure, Water Availability, Temperature
- **INTERNAL:** Crop Viability

Relationships

- **Atmospheric Pressure to Crop Viability:** POSITIVE (+) — Higher atmospheric pressure inside the habitat allows liquid water to exist and plants to photosynthesize. Below the critical threshold, water boils at room temperature and plants cannot survive.
- **Radiation Exposure to Crop Viability:** NEGATIVE (-) — Higher radiation damages plant DNA, reduces photosynthetic efficiency, and can kill crops. Effective shielding is essential for agricultural viability.
- **Water Availability to Crop Viability:** POSITIVE (+) — More available water directly supports plant growth, transpiration, and nutrient transport. Water scarcity is the most immediate threat to food production.
- **Temperature to Crop Viability:** POSITIVE (+) within range — Crop viability increases as temperature approaches the optimal growth range (18-26°C). Both extreme cold and extreme heat reduce or eliminate crop yields.

Key Discoveries

- Human survival on Mars requires simultaneously solving pressure, radiation, water, temperature, and food challenges — failure of any one system is catastrophic
- Crop viability depends on ALL four external factors being within acceptable ranges — it's the weakest link in the survival chain
- The system has almost no margin for error — a small change in any variable can cascade into colony-wide failure
- Mars colonization is fundamentally an engineering problem, not a discovery problem — we know what needs to happen, we just can't do it all yet

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mars is basically like a cold desert on Earth"	Mars is nothing like any place on Earth. Its atmosphere is 100 times thinner, there's no breathable oxygen, radiation is lethal, and the average temperature is -60°C. Even the driest, coldest desert on Earth has thousands of times more air pressure and infinitely more water than Mars.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We just need space suits to walk around on Mars"	Space suits are emergency survival equipment, not a long-term solution. They're heavy, restrict movement, require constant maintenance, and have limited life support duration. A suit breach on Mars means death in seconds. Sustainable Mars living requires permanent pressurized structures.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can terraform Mars to be like Earth"	Terraforming Mars — making it Earth-like — would take centuries to millennia if it's even possible. Mars has lost most of its atmosphere to space because it lacks a global magnetic field. Even if we could warm it, the atmosphere would slowly escape. Realistic Mars habitation means sealed habitats, not open-air living.